

I

Regional ingress and web
the way in, served locally
WAF, API edge, shell + MFE registry
proposed · no V3 ADR

N

NATS, three servers
plus a local claim resolver
a logical HA minimum, not sizing
proposed · sizing is L4

W

Workflow runtime
long-running work, cell-resident
Temporal workers and persistence
proposed · no V3 ADR

O

Stores and operations
the cell's own machinery
Postgres per owner, evidence adapter
masked logs, key handling

C

Control-plane slice
tenant lifecycle, identity, claims
holds a short-lived user credential
proposed · see LB-V3-L3-09

D

Domain services
tender, loads, fleet, documents
holds a short-lived user credential
disputes and billing sit here too

A

Integration adapters
telematics, ERP, maps, payments
holds a short-lived user credential
one connection per tenant served

K

Workflow workers
they act for one tenant at a time
holds a short-lived user credential
proposed · no V3 ADR

J

JetStream streams
the durable facts, replayable
named inside the tenant account
stream design is LB-V3-L3-05

V

NATS KV projections
read models, fast lookup, watch
a KV bucket is really a stream
so it obeys every account rule

P

Service Postgres
transactional truth, per service
tenant-scoped, and cell-resident
ownership is LB-V3-L3-07

E

Evidence documents
proof of delivery, photos, papers
S3-compatible adapter in the cell
proposed · no V3 ADR

1

An account is not a vote
3 JetStream accounts, 1 meta group
a wall for data, not for availability
measured · demo 03, T2

2

Export is the only door
one direction, subject only
read-only, and renamed on arrival
measured · demo 03, E

3

A consumer follows its stream
it lives where the stream lives
a far replica pays the long link
measured · demo 03, T2

4

Placement follows the client
a new stream lands where you are
unless a cluster is named for it
measured · demo 03, T2 and T3

?

L3-01-O01 · one credential per account, or one replica per account?
the biggest open question this sheet raises

a shared replica serving two accounts needs two connections
a replica per account is simpler, and costs far more
open · architecture to confirm · no V3 ADR

!

A replica borrows a user · it never owns an account
the cell is a place; the account is a who

a replica that held an account would make the cell a tenant
that is the mistake sheet 1 exists to stop
directional position as at 2026-09-17

SCOPE

One square, opened. What runs inside one cell, and whose account owns what it writes.

DIRECTION

Directional as at 2026-09-17, except the bottom band. That band is measured.

CELL

The cell owns the machinery. Ingress, NATS, workflow and stores exist **once per cell**.

REPLICA

A replica is a **place-thing**. It runs in a cell. It borrows a user; it never owns an account.

CREDENTIAL

One connection carries **one** user credential. Two accounts means two connections.

DATA

Streams, KV, Postgres and documents are **tenant-owned** and stay inside the cell that holds them.

KV

A KV bucket is really a stream. So it obeys every account rule a stream obeys.

HA

Three NATS servers is a **logical minimum**, not a sizing number. Sizing is L4.

MEASURED

Only the four solid tiles are measured. Everything else is proposed or open.

OPEN

L3-01-O01 is the one to answer first. It changes how every service is deployed.

Read it downward. The cell provides the machinery, a replica borrows one user credential, and everything the replica writes lands inside that user's account. **Built today:** nothing in the top three bands. The bottom band is measured — demo 03, 73 checks passed and 0 failed on 2026-09-17. **Not shown:** subjects, streams and grants, which are LB-V3-L3-05 ; process packaging and scaling, which are LB-V3-L3-12 ; claim text and permission grants, which are LB-V3-L3-09 and L4 .

SCOPE	One square, opened. What runs inside one cell, and whose account owns what it writes.
DIRECTION	Directional as at 2026-09-17, except the bottom band. That band is measured.
CELL	The cell owns the machinery. Ingress, NATS, workflow and stores exist once per cell .
REPLICA	A replica is a place-thing . It runs in a cell. It borrows a user; it never owns an account.
CREDENTIAL	One connection carries one user credential. Two accounts means two connections.
DATA	Streams, KV, Postgres and documents are tenant-owned and stay inside the cell that holds them.
KV	A KV bucket is really a stream. So it obeys every account rule a stream obeys.
HA	Three NATS servers is a logical minimum , not a sizing number. Sizing is L4.
MEASURED	Only the four solid tiles are measured. Everything else is proposed or open.
OPEN	L3-01-001 is the one to answer first. It changes how every service is deployed.

- **Cyan boundary** — what the cell owns: a place
- **Solid tile** — measured in the lab, with the run named
- **Cyan dotted line** — the cell hosts what is below it
- Line label** — names the relationship, never bare "uses"

-  **Amber boundary** – what the tenant account owns
-  **Dashed amber tile** – open, a named owner must confirm
-  **Amber dashed line** – a write landing inside an account

- cell** one region's complete runtime
- replica** one running copy of a service
- account** the NATS tenancy boundary
- user** **credential** a short-lived NATS identity
- claim** **resolver** serves account claims to a server
- JetStream** NATS durable storage and replay
- stream** a durable, replayable record of facts
- consumer** a reader's cursor over a stream
- KV** NATS key-value, a fast lookup store

- projection** a read model built from a stream
- placement** which cluster stores a new stream
- meta group** the JetStream vote across servers
- export, import** the only path between accounts
- WAF** web application firewall
- MFE** micro-frontent, a page part loaded at run time
- ERP** enterprise resource planning system
- HA** high availability
- ADR** architecture decision record

Which Account Does Each Service Connect As

Sheet 2 says a replica borrows a user credential. This sheet says **whose**. Every service named on LB-V3-L2-01 is a row; the three account types from sheet 1 are the columns. One connection sees exactly one account, so a service that needs two accounts needs two connections. **This is a directional document** as at 17 September 2026. Nothing here is built. The service names are carried down from LB-V3-L2-01 and are not restated or extended; subject and grant text belongs to LB-V3-L3-05; permission claims belong to LB-V3-L3-09 and L4.

SERVICES, DOWN · ACCOUNT TYPES, ACROSS — ONE CONNECTION SEES EXACTLY ONE ACCOUNT				
services, down accounts, across	\$SYS · administration	PLATFORM · shared corpus	TENANT · one per tenant	
CONTROL PLANE SERVICES · THEY RUN THE PLATFORM, NOT A TENANT'S BUSINESS				
Tenant + organisation lifecycle of a tenant and its orgs	not used	Connects owns tenant and org records	Imports tenants import lifecycle events	
Identity + membership who a person is, and where they belong	not used	Connects owns person and membership records	Open 007 · platform, or per tenant?	
NATS provisioning accounts, credentials, claims, placement	Connects pushes account claims · system API	Connects reads the placement record	Refused it never acts as a tenant	
Reference + config the shared corpus every tenant reads	not used	Connects owns the one shared corpus	Imports every tenant imports it, read-only	
Entitlements + admin what a tenant has bought and may use	not used	Connects owns entitlement records	Imports a tenant reads its own entitlements	
Observe + audit health, telemetry and the audit trail	Connects reads server and JetStream health	Connects writes platform audit	Open 007 · platform, or per tenant?	
DOMAIN PLANE SERVICES, ADAPTERS AND WORKERS · THEY ACT FOR ONE TENANT				
Marketplace + tender offers, bids and award	Refused a domain service is not an admin	Imports reference data, read-only · 008	Connects acts as one tenant at a time	
Loads + trips + routes the movement of goods	Refused a domain service is not an admin	Imports reference data, read-only · 008	Connects acts as one tenant at a time	
Fleet + tracking vehicles, drivers and position	Refused a domain service is not an admin	Imports reference data, read-only · 008	Connects acts as one tenant at a time	
Documents proof of delivery and paperwork	Refused a domain service is not an admin	Imports document type catalogue, read-only	Connects acts as one tenant at a time	
Disputes + acceptance claims raised against a trip	Refused a domain service is not an admin	Imports reference data, read-only · 008	Connects acts as one tenant at a time	
Billing + settlement invoices, rates and payment	Refused a domain service is not an admin	Imports currency and rate reference	Connects acts as one tenant at a time	
Integration adapters telematics, ERP, maps, payments	Refused a domain service is not an admin	Imports partner and map reference	Connects one connection per tenant served	
Workflow workers long-running work, per tenant	Refused a domain service is not an admin	Imports reference data, read-only · 008	Connects one connection per tenant served	
! Only one service ever connects as \$SYS NATS provisioning, and nothing else on this sheet every other control-plane service is a platform-account client a domain service connecting as \$SYS is refused, not merely unusual directional position as at 2026-09-17 · no V3 ADR of record		? L3-01-008 · import, or a second connection? a domain service needs tenant data and shared data one connection sees one account, so it cannot see both an import is cheaper; a second connection is simpler to reason about open · architecture to confirm · no V3 ADR		

SCOPE

Directional as at 2026-09-17. Nothing on this sheet is built, and no V3 ADR exists.

SOURCE

Every service name is carried down from LB-V3-L2-01. This sheet adds no service of its own.

RULE

One connection sees one account. A service that needs two accounts needs two connections.

SYSTEM

Only NATS provisioning connects as \$SYS . It pushes account claims. That is the whole job.

CONTROL

The control plane is platform-shaped. It owns records about tenants, not records of a tenant.

DOMAIN

The domain plane is tenant-shaped. All eight rows have the same answer, which is the point.

IMPORT

An import is read-only and renamed on arrival. Demo 03 measured that. It is the only door.

REFUSED

A red square is a decision. A domain service must never hold administrative rights.

OPEN

L3-01-007 and L3-01-008 are raised by this sheet. Both need architecture to confirm.

Read a row for a service and a column for an account. Blue means the service opens a NATS connection as that account. Grey means it only sees that account's data through an export and import, with no connection of its own. Red means the combination is refused. The shape is the finding: the control plane is platform-shaped, the domain plane is tenant-shaped, and exactly one service touches \$SYS . Built today: nothing on this sheet. Not shown: the subject and grant text, which is LB-V3-L3-05 ; permission claims, which are LB-V3-L3-09 and L4 .

SCOPE	One question. Whose credential does each service hold? Sheet 2 asks it; this sheet answers it.
DIRECTION	Directional as at 2026-09-17. Nothing on this sheet is built, and no V3 ADR exists.
SOURCE	Every service name is carried down from LB-V3-L2-01. This sheet adds no service of its own.
RULE	One connection sees one account. A service that needs two accounts needs two connections.
SYSTEM	Only NATS provisioning connects as \$SYS . It pushes account claims. That is the whole job.
CONTROL	The control plane is platform-shaped. It owns records about tenants, not records of a tenant.
DOMAIN	The domain plane is tenant-shaped. All eight rows have the same answer, which is the point.
IMPORT	An import is read-only and renamed on arrival. Demo 03 measured that. It is the only door.
REFUSED	A red square is a decision. A domain service must never hold administrative rights.
OPEN	L3-01-007 and L3-01-008 are raised by this sheet. Both need architecture to confirm.

LEGEND

- Blue boundary — a group of service replicas
- Amber panel — a rule that governs the whole sheet
- Red square, "Refused" — the platform must never allow it

- Dashed grey tile — proposed for V3, not built
- Blue square, "Connects" — opens a connection as that account
- "not used" — no relationship, and none is needed

- Dashed amber tile — open, a named owner must confirm
- Grey square, "Imports" — sees the data, holds no connection
- Column header — an account type carried from sheet 1

- Red square or panel — a combination that is refused
- Amber square, "Open" — the answer is not decided yet
- Row label — a service carried down from LB-V3-L2-01

WORDS ON THIS SHEET

- account the NATS tenancy boundary
- \$SYS the NATS system account
- platform account holds data about every tenant
- tenant account holds one tenant's own records
- connects as opens a NATS link with that account's user
- import read-only sight of another account's subjects
- export the grant that makes an import possible
- claim the signed statement that defines an account
- system API the subjects only \$SYS may speak on
- placement record names a tenant's permitted cells
- control plane services that run the platform itself
- domain plane services that run a tenant's business
- ERP enterprise resource planning system
- ADR architecture decision record